

Virtual Clinical Sandbox — Lab Activity Manual

Detailed Setup, Downloads & Hour-by-Hour Instructions for Every Week's 3-Hour Lab

This manual gives students exactly what to install/sign up for and what to do, hour by hour, in every week's 3-hour Virtual Clinical Sandbox (VCS) lab session. Distribute this alongside the PPT_Lab decks — the slides show the *what*, this document shows the *how*.

Standing safety rule: Every tool below works with synthetic or fully de-identified data only. Never upload real patient information to any of these platforms.

🔑 ONE-TIME SETUP (Complete Before Week 1)

Do these once at the start of the course — most tools are reused across multiple weeks.

#	Tool	What It's For	How to Get It
1	Synthea	Generates fully synthetic patient records	Download from github.com/synthetichealth/synthea (requires Java 11+), OR skip installation and use pre-generated sample datasets at synthea.mitre.org/downloads
2	PhysioNet account + MIMIC-CXR/MIMIC-IV access	De-identified real ICU/radiology data (used Week 5)	(a) Create a free account at physionet.org ; (b) complete the required CITI "Data or Specimens Only Research" training course at citiprogram.org ; (c) sign PhysioNet's Credentialed Health Data Use Agreement; (d) request access to MIMIC-CXR and MIMIC-IV. Approval can take several days — start this in Week 1, not Week 5.
3	Sandbox EHR (OpenEMR)	Practice EHR environment	No-install option: use the public demo at demo.openemr.io . Self-host option: download from open-emr.org
4	GNU Health	Alternate sandbox EHR / nursing care-planning module	Live demo available at gnuhealth.org ; self-hosted option also available
5	Ambient scribe trial account	Note-generation practice	Sign up for a free-tier/trial account with one vendor (e.g., Nabla, Suki, Abridge, or Freed) — pick one and stay with it all course for consistency. Confirm the trial tier does not require real patient use
	Video		Whatever your institution already uses

6	conferencing (breakout rooms)	Roleplay pairs, OSCE stations	(Zoom, Teams, Google Meet) — confirm breakout-room capability is enabled
7	Anonymization toolkit (optional but recommended)	Week 11's anonymization exercise	ARX Data Anonymization Tool, free at arx.deidentifier.org

■ CHAPTER 1 — Admin Time Savings & the AI Foundation

Week 1 Lab: Setup + Baseline Timing

Tools needed: Synthea (or sample dataset), sandbox EHR, ambient-scribe trial account, a stopwatch/phone timer.

- **Hour 1:** Create/confirm all four accounts from the one-time setup above, live and screen-shared with the instructor so no one gets stuck. Download 3–5 sample Synthea patient records to your own folder.
- **Hour 2:** Pick ONE Synthea patient. Manually write a full clinical note for a mock visit based on that patient's record — **do not use any AI tool**. Time yourself from start to finish and write the number down.
- **Hour 3:** Submit your baseline time and your handwritten note to the instructor (screenshot or upload). This number is your personal comparison point for Week 4 — save it somewhere you won't lose it. **Deliverable:** One manually-written note + your baseline time (minutes).

Week 2 Lab: Prompt Roleplay

Tools needed: Sandbox EHR, ambient-scribe trial account, video conferencing (breakout rooms), a Synthea patient card.

- **Hour 1:** Pair up in breakout rooms. One student plays the patient using a Synthea case card as their “script,” the other plays the clinician.
- **Hour 2:** Run the visit. The “clinician” writes a first prompt to the ambient-scribe tool, reviews the output, then rewrites the prompt once based on what it got wrong (apply the Four Rs).
- **Hour 3:** Submit both prompt versions side by side (a simple text document is fine). Groups reconvene for a live review of 3–4 pairs' submissions.
Deliverable: Two prompt versions (before/after) submitted as one document.

Week 3 Lab: Full Mock-Clinic Day

Tools needed: Sandbox EHR, ambient-scribe trial account, 3–4 Synthea patient

cards of increasing complexity, a spreadsheet or text file for your “catch log.”

- **Hour 1:** Run encounter 1 (simple, single-complaint visit) through the ambient scribe. Log every correction you make in your catch log, tagged as substitution/deletion/insertion.
- **Hour 2:** Run encounters 2-3 (moderate complexity). Keep logging.
- **Hour 3:** Run encounter 4 (multi-problem visit). Total your catch log — how many of each error type did you find? **Deliverable:** Completed catch log (spreadsheet or document) with error-type tags.

Week 4 Lab: Graded Checkoff

Tools needed: Sandbox EHR, ambient-scribe trial account, your Week 1 baseline time and Week 3 catch log.

- **Hour 1:** Instructor briefs the checkoff format. Review your own baseline and catch-log numbers before starting.
- **Hour 2:** Independently run one full ambient-scribed encounter (new Synthea patient), correct the note, and sign it. This is graded — no peer help.
- **Hour 3:** Submit your signed note + a short written comparison: your Week 1 baseline time vs. your Week 3-4 experience. **Deliverable:** Signed corrected note + baseline comparison writeup. **This is Checkoff 1 — pass/fail gate to Chapter 2.**

CHAPTER 2 — AI-Driven Diagnostics & the Safety Override

Week 5 Lab: Diagnostic AI Observation

Tools needed: Your approved PhysioNet/MIMIC-CXR access (confirm this went through — apply in Week 1 if not!), any free diagnostic-AI demo tool that accepts de-identified images (or an instructor-provided notebook/interface), a spreadsheet for logging.

- **Hour 1:** Download 8-10 sample de-identified chest X-ray images from MIMIC-CXR. Load them into the diagnostic-AI demo tool provided by your instructor.
- **Hour 2:** For each image, record the AI’s stated confidence score and its predicted finding in your spreadsheet — **do not act on the output yet, this week is observation only.**
- **Hour 3:** Compare the AI’s predictions against the known/reference finding for each case (your instructor will provide the answer key at the end of the session). Tally how often high confidence coincided with a wrong answer. **Deliverable:** Completed confidence-vs-correctness spreadsheet.

Week 6 Lab: Pillar Case-Sorting Exercise

Tools needed: Instructor-provided packet of 8–10 de-identified CDSS malfunction write-ups (distributed in class), video conferencing for breakout groups.

- **Hour 1:** In small groups, read through all case write-ups.
- **Hour 2:** Sort each case by which German quality pillar was violated (Functionality / Robustness / Comprehensibility) and write one sentence justifying each classification.
- **Hour 3:** Present your group's sorting to another group; reconcile any disagreements as a class discussion. **Deliverable:** Sorted case list with one-sentence justifications per case.

Week 7 Lab: Seeded Override Exercise

Tools needed: Sandbox EHR, diagnostic-AI demo tool (instructor will pre-load 2–3 seeded cases into your account before this session — you won't know which ones).

- **Hour 1:** Work through your assigned case queue in the diagnostic-AI tool as if it were a normal shift.
- **Hour 2:** For any case that seems clinically implausible, document a manual override: what the AI said, why you disagree, and what you're doing instead.
- **Hour 3:** Class debrief — students who caught a seeded case explain what tipped them off. **Deliverable:** Override documentation for any case you flagged (submit even if you flagged zero — that's useful data too).

Week 8 Lab: Graded Checkoff

Tools needed: Same diagnostic-AI demo tool, one new seeded case (instructor-provided, individual to you).

- **Hour 1:** Instructor briefs the checkoff format.
- **Hour 2:** Independently work your seeded case, observed live (camera on, or screen-share with instructor). Detect the anomaly, articulate your reasoning aloud, document the override.
- **Hour 3:** Instructor feedback session, one-on-one. **Deliverable:** Override documentation + live observation. **This is Checkoff 2 — pass/fail gate to Chapter 3.**

▣ CHAPTER 3 — Clinical Care Pathways & Data Sovereignty

Week 9 Lab: Care-Plan Drafting & Bedside Revision

Tools needed: Sandbox EHR or GNU Health, a Synthea patient record, video conferencing (breakout rooms) for roleplay.

- **Hour 1:** Generate an AI-drafted care plan from your assigned Synthea patient using the sandbox EHR's built-in NLP/care-planning feature (or instructor-provided tool).
- **Hour 2:** Pair up; one plays the patient at bedside (improvising slightly different details than the chart), the other is the nurse who must revise the care plan against what they hear.
- **Hour 3:** Submit both the original AI draft and your revised plan, with a one-line note on what changed and why. **Deliverable:** Original + revised care plan, with change rationale.

Week 10 Lab: Data-Flow Mapping

Tools needed: Any diagramming tool (draw.io/diagrams.net is free and browser-based, or plain paper/whiteboard photographed), list of every VCS tool used so far this term.

- **Hour 1:** List every tool used in Weeks 1–9 (Synthea, sandbox EHR, ambient scribe, diagnostic-AI demo, etc.) and note where each one's processing actually happens (local device, vendor cloud, etc. — ask the instructor if unsure).
- **Hour 2:** Draw a data-flow diagram showing where patient data would go at each step, in a real (non-simulated) deployment. Mark every point data would leave institutional control.
- **Hour 3:** Small-group presentation of your map to the full class. **Deliverable:** Data-flow diagram (image or PDF export).

Week 11 Lab: Anonymization & Peer Re-Identification

Tools needed: A batch of Synthea records (provided or self-generated), ARX Data Anonymization Tool (arx.deidentifier.org, free download) or instructor-approved alternative.

- **Hour 1:** Load your Synthea batch into ARX. Apply generalization/suppression to remove or generalize identifying fields (name, exact DOB, ZIP, etc.).
- **Hour 2:** Document every step you took (which fields, what transformation, why) in a text file.
- **Hour 3:** Swap your anonymized extract with a classmate. Try to re-identify a record from THEIR extract. Report whether you succeeded and how.
Deliverable: Anonymized extract + documentation + peer re-identification result (yours and your partner's).

Week 12 Lab: Graded Checkoff

Tools needed: ARX (or approved alternative), a new Synthea batch (instructor-

provided).

- **Hour 1:** Instructor briefs the checkoff format.
- **Hour 2:** Independently anonymize the new batch, documenting every step as you go.
- **Hour 3:** Submit your anonymized extract + documentation for grading.
Deliverable: Anonymized extract + full documentation. **This is Checkoff 3 — pass/fail gate to Chapter 4.** (*Consider a scheduled break after this week if your institution observes a winter recess.*)

▣ CHAPTER 4 — Ethicolegal Autonomy & the Final Checkoff

Week 13 Lab: Liability Case Conference

Tools needed: Instructor-provided de-identified adverse-event report (distributed in class), video conferencing for breakout groups, a simple diagramming tool (optional).

- **Hour 1:** Read the case report individually.
- **Hour 2:** In small groups, map the liability chain (clinician / vendor / institution) and assign proportional responsibility with justification.
- **Hour 3:** Groups present and compare — expect disagreement on the exact split. **Deliverable:** Group liability map with written justification.

Week 14 Lab: Risk-Tier Classification

Tools needed: List of every VCS tool used this term, instructor-provided summary of EU AI Act Annex III criteria.

- **Hour 1:** Individually classify each VCS tool (scribe, diagnostic-AI demo, care-planning tool) by likely risk tier, with written justification citing specific criteria.
- **Hour 2:** Pair up and cross-check each other's classifications; resolve disagreements by pointing to the specific Annex criteria, not by vote.
- **Hour 3:** Submit your final classification table. **Deliverable:** Risk-tier classification table with citations to specific criteria.

Week 15 Lab: Disclosure Roleplay

Tools needed: Video conferencing (breakout rooms), a simple grading rubric (instructor-provided).

- **Hour 1:** Instructor models one disclosure conversation live.

- **Hour 2:** In pairs, roleplay: one plays a patient who directly asks “was AI involved in my diagnosis?” The other delivers the disclosure. Swap roles.
- **Hour 3:** Peer-grade each other against the rubric (plain disclosure, verification role explained, questions invited). **Deliverable:** Completed peer-grading rubric for both roleplay directions.

Week 16 Lab: Capstone Dry Run (Ungraded)

Tools needed: All VCS tools used this term, in combination — sandbox EHR, ambient scribe, diagnostic-AI demo.

- **Hour 1:** Instructor walks through the full capstone rubric so there are no surprises later.
- **Hour 2-3:** Run one full simulated shift (intake → diagnosis → treatment planning) solo, with the instructor available for real-time feedback throughout — this is a dry run, not graded. **Deliverable:** None graded — this is a practice run; instructor feedback is verbal/live.

Week 17 Lab: Graded Checkoff

Tools needed: Same as Week 16, plus the disclosure rubric from Week 15.

- **Hour 1:** Final review, questions.
- **Hour 2:** Graded mock disclosure conversation (observed).
- **Hour 3:** Graded risk-tier classification task on a NEW tool not previously classified in class (tests transfer, not memorization). **Deliverable:** Disclosure conversation + risk-tier classification, both graded. **This is Checkoff 4 — pass/fail gate to Chapter 5.**

■ CHAPTER 5 — Consolidation & Mock OSCE

Week 18 Lab: Integration Station Design

Tools needed: Any VCS tools relevant to your group’s chosen combination, video conferencing (breakout rooms).

- **Hour 1:** In small groups, choose ≥ 2 competency areas to combine (scribing error, diagnostic override, sovereignty documentation, consent disclosure) and outline a test case.
- **Hour 2:** Build out the full station: the case details, the “trap(s),” and a simple scoring rubric.
- **Hour 3:** Pilot your station on a different peer group; collect their feedback.

Deliverable: Written station design (case + rubric) + peer feedback notes.

Week 19 Lab: Mock OSCE Round 1 (Graded)

Tools needed: Video conferencing platform with breakout/station capability, all relevant VCS tools.

- **Hour 1-3:** Timed rotation through OSCE stations (format depends on cohort size — instructor will provide the rotation schedule). Strict time limits apply per station. **Deliverable:** None submitted by student — instructor scores live against the station rubric.

Week 20 Lab: Mock OSCE Round 2 + Remediation

Tools needed: Same as Week 19.

- **Hour 1-3:** Split by need: students requiring remediation repeat specific stations; everyone else receives individual 1:1 feedback debriefs on Round 1 performance. **Deliverable:** None submitted — remediation students are re-scored; others receive verbal feedback.

Week 21 Lab: Free Practice + Readiness Sign-Off

Tools needed: Any VCS tool the student wants to revisit.

- **Hour 1-2:** Open, self-directed sandbox practice — revisit whichever tool/skill you feel least confident in.
- **Hour 3:** Individual readiness sign-off meeting with instructor. **Deliverable:** None formal — this is a readiness checkpoint, not a graded task.

■ CHAPTER 6 — Solo Pilot Capstone & Certification

Weeks 22-23 Lab: Capstone Shifts (Two Full 6-Hour Sessions)

Tools needed: All VCS tools used across the entire course, combined. *Note: these two weeks use a different format — each is a full 6-hour shift, not a 3+3 split. Budget accordingly.*

- **Shift 1 (Week 22, 6 hrs):** Full simulated shift — intake (ambient scribing) → diagnosis (diagnostic-AI validation) → treatment planning. Graded against the Confidence Factor checklist.
- **Shift 2 (Week 23, 6 hrs):** Repeat with a different synthetic case mix. Individual debrief afterward compares both shifts for consistency.

Deliverable: Two completed, graded simulated shifts.

Week 24 Lab: Targeted Remediation (as needed)

Tools needed: Whichever specific tool relates to your named gap from the capstone debrief.

- **Open/unscheduled:** Practice only the specific skill flagged in your capstone debrief. This is intentionally narrow — not a full re-do. **Deliverable:** None formal — instructor confirms gap closure informally before Week 25.

Week 25 Lab: Final Checkoff & Certification

Tools needed: All VCS tools, per your institution's final exam format.

- **Hour 1:** Final graded checkoff (format per institutional policy).
- **Hour 2:** Course evaluation.
- **Hour 3:** Certificate ceremony — AI Competency Certificate awarded.
Deliverable: Final checkoff results. **This is the Final Certification Checkoff.**

📁 Quick Reference: All Tools & Where to Get Them

Tool	URL	Used In
Synthea	github.com/synthetichealth/synthea (or synthea.mitre.org/downloads for pre-generated data)	Weeks 1, 2, 3, 4, 9, 11, 12
PhysioNet / MIMIC-CXR / MIMIC-IV	physionet.org (requires CITI training + data use agreement — apply early!)	Week 5
OpenEMR	demo.openemr.io (no install) or open-emr.org (self-host)	Multiple weeks
GNU Health	gnuhealth.org	Week 9 (alternate)
ARX Data Anonymization Tool	arx.deidentifier.org	Week 11, 12
Ambient scribe trial (pick one)	Vendor sign-up (Nabla, Suki, Abridge, Freed, etc.)	Weeks 1-4, 16, 17, 22-23
draw.io / diagrams.net	app.diagrams.net (free, browser-based)	Week 10
Video conferencing	Institutional standard (Zoom/Teams/Meet)	Weeks 2, 6, 9, 13, 15, 18-20

All access procedures (especially PhysioNet/MIMIC credentialing) should be re-verified each term, since requirements and approval timelines can change.